Space Astrophysics (ASTR 3161)

Spring 2026

Course and Contact Information

Department and Course:	Physics, ASTR 3161
Semester:	Spring 2026
Time:	Monday and Wednesday 12:45 PM - 02:00 PM
Location:	Corcoran Hall 404A
Course Web Site:	http://blackboard.gwu.edu

Instructor:	Prof. Oleg Kargaltsev
Campus address:	Corcoran Hall 407 (office)
Phone:	202-994-7225 (office)
E-Mail:	kargaltsev@gwu.edu
Office Hours:	Wed 4-5 pm, Corcoran 407
	(or by individual appointment)

Course timeline:

Week	Date	Homework Due or Exam	Topics
1	Mon Jan 12		A Brief History of High-Energy (X-ray & Gamma-Ray) Astronomy. Basic terminology.
2	Wed Jan 14		Radiation I: Electromagnetic Radiation and Basic Radiation Processes.
2	Mon Jan 19		Martin Luther King Day (no class)
3	Wed Jan 21		Radiation II: Electromagnetic Radiation: Radiation quantities.
3	Mon Jan 26		Radiation III: Thermal equilibrium. Blackbody. Optical depth.
4	Wed Jan 28		Radiation IV: Absorption in ISM. Thermal Bremsstrahlung.
4	Mon Feb 2		Radiation V: Line emission. Cyclotron radiation.
5	Wed Feb 4	HW#1	Radiation VI: Cyclotron radiation.
5	Mon Feb 9		Radiation VII: Synchrotron Radiation.
6	Wed Feb 11		Radiation VIII: Synchrotron Radiation. Thomson scattering
6	Mon Feb 16		President's Day (no class)
7	Wed Feb 18		Compton Scattering, Inverse Compton Scattering
7	Mon Feb 23	HW#2	Basic plasma physics. MHD. Reconnection. Dynamo.
8	Wed Feb 25		Shock waves. Particle acceleration.
8	Mon Mar 2		Particle acceleration. Review for midterm #1. Final project intro.
9	Wed Mar 4		EXAM #1
9	Mon Mar 9		SPRING BREAK
10	Wed Mar 11		SPRING BREAK
10	Mon Mar 16	HW#3	Telescopes and Detectors for X-rays and, Gamma-rays.

11	Wed Mar 18		Sources of High-Energy Emission: Evolution of stars. White Dwarfs. Supernovae. Supernova Remnants.
11	Mon Mar 23		Sources of High-Energy Emission: Isolated Neutron Stars. Pulsars. Pair production.
12	Wed Mar 25		Sources of High-Energy Emission: Pulsar wind nebulae. Neutron star zoo.
12	Mon Mar 30	Final project topic is due	Accretion. White Dwarfs and Neutron Stars in Binary Systems.
13	Wed April 1		Isolated Black Holes. Accreting black holes in binaries.
13	Mon April 6		Astrophysics and observational manifestations of supermassive Black Holes, AGNs.
14	Wed April 8	Final project outline is due	Gamma-ray Bursts.
14	Mon April 13		Fast Radio Bursts.
15	Wed April 15		Gravitational waves.
15	Mon April 20	HW#4	EXAM #2
16	Wed April 22	Final project Progress Report is due on BB (10 pts extra HW credit).	. Cosmic Rays, High-Energy Particle Astrophysics, Neutrino Astrophysics.
16	Mon April 27		Practical demonstration: data from Chandra X-ray Observatory and Hubble Space Telescope. Last minute questions about final project questions
	May 4	Due at 3 pm	Submit Term Project by uploading to BB->Term Project

Detailed Course Description

COURSE DESCRIPTION: In this survey course you will learn about high-energy radiation processes, the 4th state of matter (plasma), extreme states of matter and space (neutron stars and black holes), basic astronomy and lives of stars, modern X-ray and gamma-ray observatories, and basics of statistical data analysis.

PREREQUISITE: PHYS 1022 (University Physics II) and MATH 1232 (Calculus II) or equivalent. You need to know material from these courses well to be successful in ASTR 3161 course.

LECTURES: The lectures are meant to explain difficult concepts, to expand on the reading material, and to introduce topics not covered in the textbook such as examples from real observations and data. If you do not understand something, you are very much encouraged to ask questions during the lectures. **The lecture content for the previous weeks and for the upcoming week, the reading assignments, and the homework will be listed in the Weekly Activities section of the Blackboard course page. Please check that page frequently, things may change. If needed, that page is updated announcements will be made through the Blackboard Announcements.**

YOUR RESPONSIBILITIES:

1. Reading any materials distributed or assigned otherwise;
2. Actively participating in discussions during the class;
3. Repeating the any calculations and derivations done in class alone by yourself (with textbook/notes closed);
4. Doing your homework yourself and turning it in time.

At least 6 hrs/week of independent learning is expected in this course. This will help you do well on the exams.

LEARNING OBJECTIVES:

In this course, you will:

1. Acquire basic knowledge about cosmic sources of high-energy radiation;
2. Develop analytical, scientific, and critical thinking skills;
3. Develop the ability to identify the appropriate physics laws/principles and mathematical methods needed to tackle specific astrophysics problem;
4. Make connections between mathematics, physics and astronomy.

TEXTBOOKS AND OTHER USEFUL BOOKS:

There are no required textbooks for this course. The field is developing rapidly and there are no textbooks fully covering the scope of this source.

There is one recommended book which overlaps a lot with the course material, but it was last updated in 2010.

Exploring the X-ray Universe: Second Edition
F.D. Seward and P.A. Charles
Cambridge University Press, 2010
ISBN: 978-0-521-88483-9

The lecture slides and notes distributed vial Blackboard should be sufficient to do well in this course.

However, I also strongly encourage you to look through the preprints abstracts http://arxiv.org/list/astro-ph/recent looking for articles relevant for each class. If something catches your attention we can discuss it at the beginning of the class. This will count toward your class participation.

Another good resource is NASA's database of publications: https://ui.adsabs.harvard.edu/ Where in addition to research articles you can find recent reviews on various topics related to our course.

GRADING: Your grade will be based on homework (15%), exam #1 (20%), exam#2 (20%), in-class quizzes (15%), attendance/participation (10%), term project paper (20%). Lectures are mandatory and active class participation is expected (part of the grade). If you missed any lectures because of medical reasons, you must bring a note from your doctor stating that you have been sick on that day.

The grading scale for the course is as follows:

106.00 - 94.000	A
93.999 - 90.000	A–
89.999 - 86.000	B+
85.999 - 82.000	B
81.999 - 78.000	B–
77.999 - 74.000	C+
73.999 - 70.000	C
69.999 - 66.000	C–
65.999 - 61.000	D+
60.999 - 55.000	D
54.999 - 50.000	D–
49.999 - 0.000	F

HOMEWORK AND TEAMWORK POLICY: Homework assignments will be distributed and collected **BEFORE** class starts on the **specified day** (see pages 2-3). The homework problems will also be available through the Problem Sets section of the course in the Blackboard. Any homework turned in late will be assigned only half of the credit. Please start working on your homework early and do not postpone it till the last day! If you have a medical excuse, you must contact me as soon as possible (e.g. by e-mail) to arrange a new due date for the HW.

In this class you are expected to learn how to think independently and creatively and how to solve problems by yourself. You are required to work by yourself (alone) on each problem for an hour before you discuss it with anyone or come to office hours. Feel free to consult the textbooks and lecture notes but *do not Google the solution* and paste it. I am aware of what is online and will notice it. After you have made an honest attempt to solve the homework problem for at *least 1 hour*, you may discuss it with other students or come see me during the office hours. Any substantial use of online resources must be clearly indicated in your homework. Should a particular problem cause troubles for majority of students in the class, I will review the solution at the beginning of my next lecture.

It is VERY important to write you homework solution in a detailed and clear way. If I cannot follow your solution, I cannot assign any credit for it. A detailed, well-structured solution will usually get higher score. Please, do not substitute the numbers into the equations until the very end, whenever possible. Clearly number the problems in your solution in the same way they are numbered in the homework assignment. A partial credit for the homework (or exam) problem can be earned even if you got a wrong answer but demonstrated understanding of underlying physics. Staple your homework before handing it in. Some homework sets may include challenging problems for those who plan to go on to become professional astrophysicists or would like to earn a little extra credit.

Each qualitative problem (homework or exams) will be graded in the following way: 20% is awarded if you demonstrate understanding of the physical processes associated with the problem. Another 20% is awarded if you use the correct equations (assuming equations are needed). Another 40% are awarded for the correct solution of equations. Final 20% is awarded if the numerical answer (if it is required) is correct. You may get some points if you got the wrong numerical answer but realized this and wrote why you think it is unreasonable**.**

If you disagree with your homework grade, you must e-mail your appeal to me **within 1 week** from the date when graded homework was returned to you. The appeal must include your name, it should clearly identify the issue in question, contain a detailed explanation of what you think is wrong or unfair, and your original homework must be included as a part of the appeal. The same procedure applies to the exams.

EXAMS: There will be one **two** exams in this course. **The exams will be based solely on the material covered during the lectures, assigned reading, and in the homework.** You are responsible for knowing the material presented in class as well as material from the assigned readings (including the textbook material and the distributed in class materials, unless the material is marked as optional). The exams will be closed book and closed notes (except for those distributed with the exam). You may use standard, non-

programmable calculators on the exams. Cell phones, laptops, Ipads, and calculators that can store equations and text are not allowed. A violation of this policy will cost you half the exam score. A table with physical constants (in cgs units) will be provided during the exam.

Each exam will have 4-5 problems (each problem may have multiple simpler parts). Questions may include definitions, physical explanations, brief calculations, and brief derivations. You should work efficiently to score as many points as possible. Do not get stuck with a single question for the entire exam period. Incomplete solution can get partial credit so please write at least something reasonable. To get full or partial credit, your written answers must be written legibly, and your writing must show your reasoning clearly. Please provide your solutions in the same order in the problems are given to you.

No makeup exams are planned. If you have a medical excuse, you must contact me as soon as possible regarding this matter to discuss possible solutions.

QUIZZES: There will be several brief quizzes typically given before class starts. They will be announced ahead of the class start.

FINAL PROJECT: This is your independent research project. You will need to write and submit to the instructor by the specified date a research paper (a proposal to observe a particular high-energy source or your own in-depth investigation of an astrophysical phenomenon/object relevant to the course, using available literature). Each student should hand in *a one-paragraph outline of the project*, on **Monday, April 8.**. This outline should include the goal of the project, the data to be used, and a list of six or more relevant references to the journal articles. NASA ADS service is a great place to search for articles: http://adsabs.harvard.edu/abstract_service.html. A *Progress Report will be due* on **Monday, April 22**. This is an expansion of the outline and should include your work to date, as well as any problems you are encountering with your project. ***Your final project report should include***: a cover page with a title and abstract, introduction describing the question you have posed and would like to investigate, available data and literature, brief description of any relevant analysis methods or proposed observations, tabular/graphical results, images, conclusions, suggestions for future work, and reference list. The total length of the report (including figures) should not exceed ten pages using 11pt font, double spaced. Electronic copy of your term project **must be e-mailed to me by 5 pm on the final exam date of this course** (the date is set by CCAS). If you do not understand what you should do for the final project or you are having other difficulties with your term project, you must let me know **in advance** and come to see me during the office hours. Late reports will not be accepted unless there is a valid (e.g., medical) reason.

AI (ChatGPT etc) policy: I strongly discourage you from using ChatGPT while solving homework because you will not have AI with you during the exams. However, you may use Chat GPT (or other AI) for your term project, if you find it useful. Any erroneous information provided by ChatGPT (or other AI) is not going to be accepted as a valid excuse for raising the grade.

CLASS ATTENDANCE AND PARTICIPATION: The class attendance and participation is important. To do well, you should (1) come to class and pay attention, (2) answer questions when they are posed by the instructor, (3) ask questions in class when you don't understand the lecture, (4) perform simple tasks when requested by the instructor, (5) be

courteous and friendly to your fellow students and the instructor, and (6) follow the general points on classroom conduct given below. You must also make sure that everything taught is clear to you. Please feel free to ask questions regarding the previous class material at the beginning of the next class.

GENERAL CLASSROOM CONDUCT: Please turn off cell phones and Ipads before the start of each class. No food consumption during the lecture unless required for medical reasons (must notify instructor ahead of the lecture). Please respect your classmates and do not talk during the class unless asked by the instructor.

OFFICE HOURS AND QUESTIONS: You are strongly encouraged to come to my office during the office hours for help with the course material. If you cannot make the appointed times, please e-mail to make an appointment. If you are unhappy about something in the course please let me know, I will discuss it with you and try to fix it.

University policies and resources

Academic Integrity Code

Academic integrity is an essential part of the educational process, and all members of the GW community take these matters very seriously. As the instructor of record for this course, my role is to provide clear expectations and uphold them in all assessments. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and otherwise violate the Code of Academic Integrity. If you have any questions about whether particular academic practices or resources are permitted, you should ask me for clarification. If you are reported for an academic integrity violation, you should contact Conflict Education and Student Accountability (CESA), formerly known as Student Rights and Responsibilities (SRR), to learn more about your rights and options in the process. Consequences can range from failure of assignment to expulsion from the University and may include a transcript notation. For more information, refer to the CESA website at students.gwu.edu/code-academic-integrity or contact CESA by email cesa@gwu.edu or phone 202-994-6757.

University Policy on Observance of Religious Holidays

Students must notify faculty during the first week of the semester in which they are enrolled in the course, or as early as possible, but no later than three weeks prior to the absence, of their intention to be absent from class on their day(s) of religious observance. If the holiday falls within the first three weeks of class, the student must inform faculty in the first week of the semester. For details and policy, see provost.gwu.edu/policies-procedures-and-guidelines.

Use of Electronic Course Materials and Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials

and/or recorded class sessions. Contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

Academic support

Academic Commons

Academic Commons is the central location for academic support resources for GW students. To schedule a peer tutoring session for a variety of courses visit go.gwu.edu/tutoring.
Visit academiccommons.gwu.edu for study skills tips, finding help with research, and connecting with other campus resources. For questions email academiccommons@gwu.edu.

GW Writing Center

GW Writing Center cultivates confident writers in the University community by facilitating collaborative, critical, and inclusive conversations at all stages of the writing process. Working alongside peer mentors, writers develop strategies to write independently in academic and public settings. Appointments can be booked online at gwu.mywconline.

Support For Students in and Outside the Classroom

Disability Support Services (DSS) 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services at disabilitysupport.gwu.edu to establish eligibility and to coordinate reasonable accommodation.

Student Health Center 202-994-5300, 24/7

The Student Health Center (SHC) offers medical, counseling/psychological, and psychiatric services to GW students. More information about the SHC is available at healthcenter.gwu.edu. Students experiencing a medical or mental health emergency on campus should contact GW Emergency Services at 202-994-6111, or off campus at 911.

GW Campus Emergency Information

GW Emergency Services: 202-994-6111

For situation-specific instructions, refer to GW's Emergency Procedures guide.

GW Alert

GW Alert is an emergency notification system that sends alerts to the GW community. GW requests students, faculty, and staff maintain current contact information by logging on to alert.gwu.edu. Alerts are sent via email, text, social media, and other means, including the Guardian app. The Guardian app is a safety app that allows you to communicate quickly with GW Emergency Services, 911, and other resources. Learn more at safety.gwu.edu.

Protective Actions

GW prescribes four protective actions that can be issued by university officials depending on the type of emergency. All GW community members are expected to follow directions according to the specified protective action. Learn more at safety.gwu.edu/gw-standard-emergency-statuses

Miscellaneous

***Blackboard* system:** The *Blackboard* courseware system will be used for the ASTR 3161 class. The address for the *Blackboard* web site is: **http://blackboard.gwu.edu**

After entering *Blackboard*, it is necessary for you to click on the ASTR 3161 course. You are *automatically* subscribed within the *Blackboard* system to the courses for which you are registered (but you also must have a GW e-mail address!). The web access provided by *Blackboard* is a valuable resource for all aspects of the class. It includes course announcements, lecture notes, homework solutions, discussion forums, and other useful features. You should consult it frequently!

ABOUT YOUR INSTRUCTOR: Dr. Oleg Kargaltsev is an Associate Professor in the Department of Physics. He works extensively with Chandra X-ray Observatory (NASA) the X-ray Multi-Mirror Mission-Newton (ESA), Suzaku X-ray Observatory (JAXA) and Hubble Space Telescope (NASA) to study the physics, evolution, and properties of neutron stars and black holes. He is an author of >200 research papers and Principal Investigator of many observing programs on the various space observatories. Kargaltsev's other interests include science philosophy and policy, logical (especially recursive) puzzles, and science fiction.